

FIX FRAME / PRODUCT + DESIGN + ENGINEERING

COMPLETE WEBSITE DEVELOPMENT BLUEPRINT

A production-ready guide for building the senior-level college-event videography website, dynamic admin system and enquiry workflow shown in the approved visual direction.

**DESIGN
STANDARD**

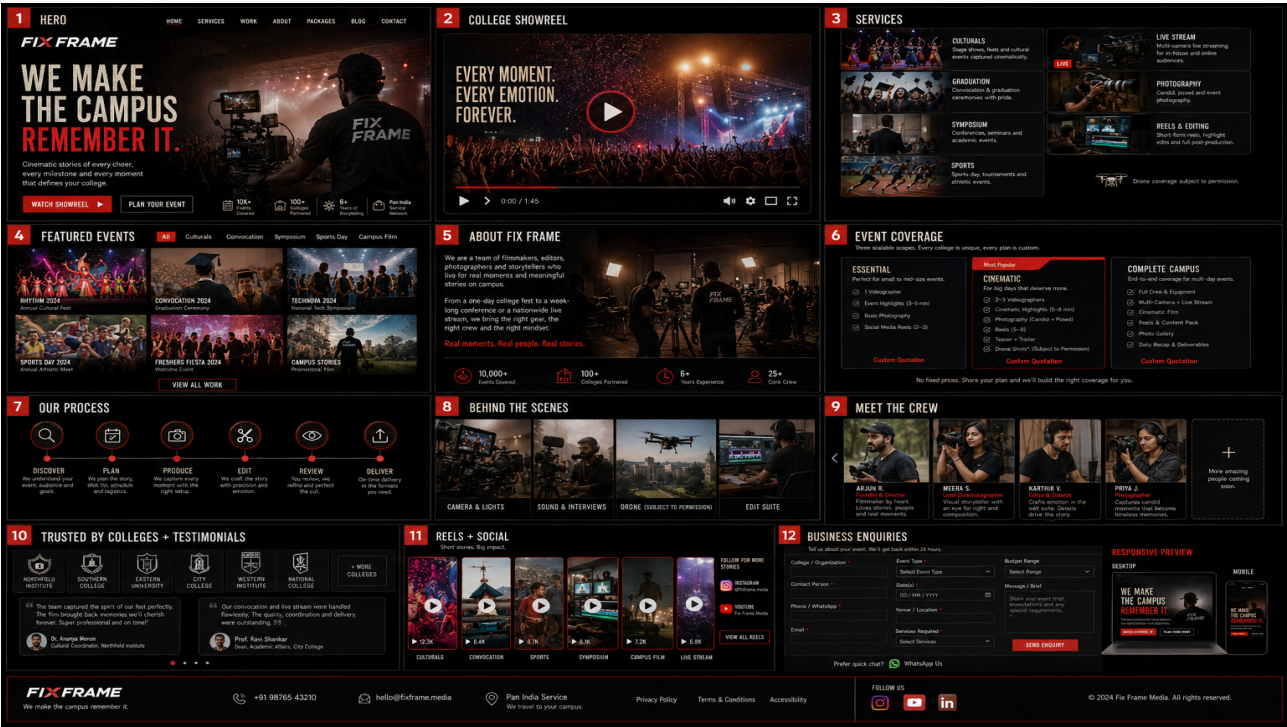
**Cinematic in presentation.
Precise in business.
Maintainable after launch.**

Public website • CMS/Admin • Media pipeline • CRM • SEO • Accessibility •
Deployment

01 / IMPLEMENTATION BLUEPRINT

Approved product direction

The visual board is the design target. The implementation must preserve its hierarchy and business clarity while replacing all placeholder data with real content.



Preserve	Improve during implementation
Dark cinematic identity	Use authentic Fix Frame and college-event media
Complete section coverage	Improve typography, exact copy and content accuracy
Visible business enquiry	Connect it to validation, database, email and CRM
Team, clients and reels	Make every collection dynamic and scalable
Readable numbered board	Translate into a coherent scrolling website, not literal panels

NON-NEGOTIABLE

The production website must not contain invented client logos, team biographies, statistics, testimonials or contact details. Placeholder content is acceptable only in development environments.

02 / IMPLEMENTATION BLUEPRINT

Core architecture decisions

These decisions create a maintainable system that fits your .NET experience and lets the owner publish normal content without developer support.

Area	Decision	Reason
Public frontend	Next.js + React + TypeScript	SEO-friendly rendering, strong image handling and responsive component model
Backend	ASP.NET Core Web API	Fits your established DTO, service, repository and validation approach
Database	Managed PostgreSQL	Reliable relational content, JSONB for controlled flexible metadata
Admin	Protected Next.js/React admin using the same API	Owner can manage content, leads and settings
Video	Cloudflare Stream	Adaptive streaming, poster/playback delivery and encoding
Media source	Cloudflare R2	Original media source-of-record and CDN-compatible object storage
Publishing	Database status + signed revalidation webhook	Content changes appear without a new deployment
CRM	Enquiry records inside the application	Structured follow-up rather than WhatsApp-only leads

CLEAR BOUNDARY

Adding or editing records requires no deployment. Creating a completely new section type, workflow or component still requires development and release.

03 / IMPLEMENTATION BLUEPRINT

Product scope and user journeys

Treat this as a small business platform with a premium marketing surface, not only as a portfolio page.

Public visitor journey

Discover work -> understand capability -> compare coverage -> trust the crew
-> submit event brief -> receive confirmation -> sales follow-up

Owner journey

Sign in -> add project/reel/client/team member -> preview -> publish
-> website revalidates -> content becomes visible without deployment

Sales journey

New enquiry -> verify date/scope -> contact -> prepare quotation
-> Quoted -> Won/Lost -> optional conversion to operational project

User	Primary goal	Success condition
College management	Find a reliable production partner	Clear proof, process and enquiry
Faculty coordinator	Define coverage and delivery	Structured scope and fast response
Student committee	See energy and social outputs	Relevant reels and event films
Site owner	Keep the website current	Publish independently
Sales user	Progress qualified leads	Visible status, ownership and notes

04 / IMPLEMENTATION BLUEPRINT

Information architecture

The homepage gives the strongest overview. Dedicated pages prevent future content growth from making it unmanageably long.

Route	Purpose	Dynamic source
/	Hero, showreel, featured services/work, process, team, trust and enquiry	Site settings + featured records
/work	Filterable project archive	Projects + project media
/work/[slug]	Case study with film, brief, approach, deliverables and results	Project record
/services	All production and post-production services	Services
/services/[slug]	Service-specific proof and enquiry CTA	Service + related projects
/packages	Coverage comparison and custom quote entry	Packages + features
/about	Studio story, process and complete team	About blocks + team
/reels	Growing vertical-video archive	Reels
/contact	Detailed business enquiry	Site settings + enquiry form
/privacy, /terms	Legal and consent information	Managed legal pages

HOMEPAGE RULE

Use Featured + DisplayOrder + PublishStatus. Never render every reel, client or team member on the homepage.

05 / IMPLEMENTATION BLUEPRINT

Senior-level visual system

The design quality comes from disciplined decisions, real media and hierarchy—not from adding more visual effects.

Foundation	Specification
Color	Ink #0A0A0A; charcoal #171717; signal red #E3262E; warm paper #F2EFE8; accessible muted text
Typography	Condensed display face for short headings; Inter-style sans for navigation, body, forms and metadata
Grid	12 columns desktop, 8 tablet, 4 mobile; max content width 1280px
Spacing	8px base with 16, 24, 32, 48, 64, 96 steps
Corners	Mostly square or 2-6px; avoid universal rounded-card styling
Borders	Thin neutral lines; red only for active state and action emphasis
Imagery	Authentic production and campus moments; consistent color grade and focal-point metadata
Copy	Short editorial headlines; plain-language deliverables and process descriptions

Professional restraint

- Use different section compositions instead of repeating the same six-card grid.
- Avoid glow, glass surfaces, floating decorative shapes and irrelevant 3D objects.
- Keep long text on warm/light surfaces when that improves reading comfort.
- Use red as an interaction signal, not as a background for entire pages.
- Design all empty, loading, error and long-content states—not only the ideal screenshot.

06 / IMPLEMENTATION BLUEPRINT

Responsive layout rules

Responsive design is a planned content transformation, not simply shrinking the desktop screenshot.

Breakpoint	Layout behavior	Important changes
$\geq 1280\text{px}$	12-column, wide editorial compositions	Full navigation, asymmetric work grids, comparison table
1024-1279px	12-column compact	Reduce media height; preserve hierarchy
768-1023px	8-column tablet	Two-column services/team; packages stack or scroll
480-767px	4-column mobile	Single-column story; filter chips; sticky enquiry action
$< 480\text{px}$	Compact 4-column	Shorter hero, simplified metadata, 20px page padding

Component rules

- Hero text remains readable without depending on the background image.
- Hover information becomes visible content or a deliberate tap interaction.
- Package comparison becomes stacked cards or an accessible table with horizontal overflow.
- Team grid automatically moves 4 → 2 → 1 columns.
- Reel archive uses responsive virtualized/paginated results when volume grows.
- Business form uses the correct mobile keyboard for phone, email and date fields.

07 / IMPLEMENTATION BLUEPRINT

Motion and interaction specification

Motion should communicate state, rhythm and craft while remaining fast and optional for users who prefer reduced motion.

Element	Default motion	Accessible fallback
Hero	Subtle image scale 1.02→1.00 and text reveal, 500-700ms	Static first frame
Showreel	Modal fade/scale and controlled player transition	Instant open; keyboard close
Project grid	150-220ms image crop and metadata reveal	Metadata permanently visible
Process	Step line activates on scroll	Static numbered timeline
Before/after	Pointer/keyboard slider	Two labeled comparison images
Navigation	Active section underline and compact sticky state	No motion required
Form	Inline validation and success-state transition	Immediate textual messages

MOTION BUDGET

Use CSS transitions where possible. Load a motion library only for interactions that cannot be implemented clearly with CSS. Respect prefers-reduced-motion everywhere.

08 / IMPLEMENTATION BLUEPRINT

Frontend component inventory

Create a controlled component system before assembling pages. This keeps the website polished as content changes.

Group	Components
Global	Header, mobile menu, footer, announcement, skip link, cookie/privacy notice
Actions	Primary button, secondary button, text link, play button, WhatsApp action
Media	Responsive image, video poster, showreel modal, reel player, gallery, comparison slider
Content	Section heading, stat, service row, project card, case-study block, process timeline
Collections	Project filter, reel grid, client logo wall, team grid, testimonial carousel/list
Commercial	Package card, comparison table, enquiry CTA, package preselection
Forms	Field, select, date, checkbox group, file/link input, error summary, success state
System	Skeleton, empty state, error state, pagination, load more, preview badge

Every component needs

- Desktop, tablet and mobile behavior.
- Default, hover, focus, active, disabled, loading and error states.
- Content-length testing with short and long real-world values.
- Keyboard behavior and accessible name.
- Analytics event only where it supports a business decision.

09 / IMPLEMENTATION BLUEPRINT

Homepage implementation map

Each section should have a content query, visual component and measurable business purpose.

Section	Data rule	Primary event
Hero	Active hero + optional showreel	hero_cta_clicked
Showreel	Current featured showreel	showreel_started / completed
Services	Active services ordered by DisplayOrder	service_viewed
Featured events	Published + IsFeatured; maximum 6	project_opened
About	Published story block + approved statistics	about_opened
Packages	Active packages ordered; no false pricing	package_selected
Process	Managed steps or stable configuration	process_viewed
BTS	Featured media collection	bts_opened
Team	Published + IsFeatured; maximum 4	team_opened
Clients/testimonials	Approved + permission valid	testimonial_viewed
Reels	Published + featured; maximum 6	reel_started
Enquiry	Form configuration + site settings	enquiry_submitted

RENDER STRATEGY

Server-render business-critical content. Use client components only for filters, media controls, motion and forms. Do not turn the entire homepage into a client-side application.

10 / IMPLEMENTATION BLUEPRINT

Work archive and case studies

Projects are the strongest sales proof. They need structured stories, not only a gallery of attractive frames.

Project record

- Title, slug, college/organization, event type, date, city and venue.
- Brief, challenge, approach, deliverables and result.
- Cover image, showreel/video, gallery, credits and related services.
- Audience size, cameras, crew, turnaround and verified metrics when permitted.
- SEO title, description, social image, canonical and indexing control.
- Status, publish date, featured flag and display order.

Archive behavior

- Server-readable filter URLs such as `/work?type=graduation`.
- Pagination or Load More; never request the entire media library.
- Keep filter state in the URL for sharing and browser navigation.
- Project pages include a next relevant project and an enquiry CTA with `sourceProjectId`.
- If a project is unpublished, return 404 publicly but preserve it in admin preview.

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Services, packages, about and team

These sections must remain flexible without allowing the admin to damage the design system.

Content	Owner controls	System controls
Services	Name, summary, inclusions, image, related work, order	Template, typography and responsive layout
Packages	Name, best-for, feature rows, price label, active state	Maximum recommended tiers and comparison behavior
About	Story blocks, proof points and selected images	Approved block types and spacing
Team	Name, role, bio, portrait, links, featured/order	Crop, responsive grid and homepage limit
Clients	Name, logo, URL, permission and featured/order	Logo normalization and archive behavior

DESIGN PROTECTION

The owner manages structured fields—not arbitrary fonts, colors, animation values or freeform page-builder positioning. New content inherits the professional component system automatically.

Growth example

If the team grows from four to twelve members, the homepage still shows the selected four. The About/Team page displays all twelve using automatic pagination or a responsive grid. No deployment is required.

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Business enquiry implementation

The form must qualify the event, work on mobile and create a reliable business record.

Step	Fields
Contact	College/organization, contact person, designation, phone, email, preferred contact
Event	Type, title, date, start/end time, venue, city, number of days, audience estimate
Coverage	Video, photo, reels, interviews, drone, live stream, same-day edit
Deliverables	Highlight length, reel quantity, complete recordings, photos, deadline
Commercial	Budget range, reference link, message, consent
Attribution	sourceProjectId, packageId, serviceId, campaign and landing page

Submission sequence

```
Validate client + server -> rate limit -> bot check -> idempotency check
-> save enquiry -> save source attribution -> email sales
-> show reference number -> optional WhatsApp deep link
```

DO NOT LOSE LEADS

WhatsApp is a convenience action, not the system of record. A successful structured submission must be stored before redirecting or offering WhatsApp.

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Admin application structure

The admin should be simple enough for the owner but controlled enough to protect content quality and business data.

Screen	Capabilities
Dashboard	Lead summary, publishing queue, failed media, expiring rights, recent activity
Projects	Create/edit, gallery, video, related services, SEO, preview and publish
Reels	Upload/link, poster, caption, tags, featured, order and scheduling
Media	Originals, derivatives, processing status, alt text, focal point and rights
Team/clients	Profiles/logos, permissions, active, featured and ordering
Services/packages	Structured features, active state, order and quotation preselection
Testimonials	Submit, approval, permission, associated project and display control
Enquiries	Status, owner, notes, activity, source and export
Pages/SEO	Global settings, metadata, legal pages, redirects and navigation
Users/roles	Invite, deactivate and permission composition

Admin UX rules

- Autosave drafts but require explicit Publish.
- Show image/video processing status and actionable failures.
- Provide Preview using an expiring signed token.
- Warn before unpublishing content linked from navigation or campaigns.
- Never expose raw storage credentials to the browser.

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Dynamic publishing model

This is the mechanism that allows content growth without code edits or redeployment.

```
Admin saves Draft
-> API validates structured fields
-> media reaches Ready status
-> owner previews
-> owner publishes
-> API commits PublishStatus + PublishedAt
-> signed webhook calls Next.js revalidation endpoint
-> cache tags for affected pages are refreshed
-> visitors see the new record
```

Collection	Homepage rule	Full-page rule
Projects	Featured, maximum 6	Paginated/filterable archive
Reels	Featured, maximum 6	Paginated archive
Team	Featured, maximum 4	All active members
Clients	Featured logo set	All approved clients
Testimonials	Featured, maximum 2-4	Contextual lists
Packages	All active within recommended limit	Full comparison

NO-DEPLOYMENT GUARANTEE

Normal create, edit, reorder, feature, publish, unpublish and archive actions do not require a frontend build. Schema or component changes do.

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Database model

Use relational entities for business integrity and JSONB only for carefully bounded metadata that does not justify a separate table.

Entity	Core relationships / purpose
Project	EventType, services, media, client, testimonials, SEO
MediaAsset	Original object, derivatives, dimensions, focal point, alt text, status
VideoAsset	R2 source, Stream UID, duration, playback status, captions, poster
Reel	VideoAsset, caption, tags, event type, featured/order
Service	Projects, package features, enquiry preselection
Package / Feature	Ordered comparison rows and active scope
TeamMember	Role, bio, portrait, active/featured/order
Client	Logo, permission, projects and testimonials
Testimonial	Client, person, designation, approval and permission
Enquiry	Contact, event, scope, attribution, status, assigned user
EnquiryActivity	Status change, note, actor and timestamp
SeoMetadata	Title, description, canonical, social image, index state
MediaRight	Consent/license source, territory, expiry and usage scope
SiteSetting	Contact, social, service area and global content

Common columns

Id, Status, CreatedAt, CreatedBy, UpdatedAt, UpdatedBy, RowVersion. Publishable records additionally use Slug, PublishStatus, PublishedAt, ScheduledAt, IsFeatured and DisplayOrder.

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API surface and conventions

Keep public reads optimized and admin writes explicit. Do not expose generic unrestricted CRUD endpoints to the public site.

Method	Endpoint	Purpose
GET	/api/public/home	Aggregated homepage payload
GET	/api/public/projects	Published archive with filters/pagination
GET	/api/public/projects/{slug}	Published case study
GET	/api/public/services/{slug}	Service and related work
POST	/api/public/enquiries	Validated idempotent lead creation
POST	/api/admin/media/uploads	Create authorized upload session
POST	/api/admin/projects	Create draft project
PUT	/api/admin/projects/{id}	Update with concurrency guard
POST	/api/admin/projects/{id}/publish	Publish and revalidate
POST	/api/admin/content/reorder	Transactional display ordering
PATCH	/api/admin/enquiries/{id}/status	Lead progression
POST	/api/webhooks/stream	Verify and update video processing

Conventions

- FluentValidation for command-specific rules.
- Idempotency-Key for enquiry and upload/publish commands.
- Cursor or page-based pagination with stable ordering.
- ETag/row-version concurrency for admin edits.
- ProblemDetails errors with trace ID; never leak internal exceptions.
- OpenAPI contract and generated TypeScript client where practical.

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Media upload and processing pipeline

Video and photography are the product. The pipeline must preserve originals, create efficient derivatives and surface failures clearly.

Video

```
Admin requests upload -> API creates MediaAsset(Pending)
-> browser uploads original to R2 with signed URL
-> API verifies object -> queue/job submits source to Stream
-> Stream webhook -> Ready/Failed -> store UID, duration and playback data
-> owner selects poster/captions -> publish allowed only when Ready
```

Images

```
Upload original to R2 -> validate MIME/signature/dimensions
-> strip unsafe metadata -> create AVIF/WebP/JPEG derivatives
-> store width/height/focal point/alt text -> CDN delivery
```

Required safeguard	Rule
File validation	Check signature, MIME, size, dimensions and authorized extension
Source of record	Preserve original in R2 before downstream derivatives
Rights	Block publication if required consent/license is missing or expired
Deletion	Soft-delete first; purge derivatives through controlled retention job
Failure handling	Retry bounded transient failures; expose final actionable status

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Authentication, roles and permissions

Compose permissions into roles so access remains precise when additional staff join the business.

Role example	Typical permissions
Owner	All content, enquiries, users, settings, publish and audit access
Content editor	Create/edit projects, reels, team and media; preview; no user management
Publisher	Review and publish approved content
Sales	Read/update enquiries and notes; read packages; no media deletion
Viewer	Read-only dashboard and records

Permission examples

```
projects.read  projects.write  projects.publish
media.upload   media.delete   testimonials.approve
leads.read     leads.assign   leads.export
settings.write users.manage    audit.read
```

- Use secure HttpOnly cookies for the same-site admin where possible.
- Require MFA for Owner and Publisher.
- Log sign-in, publish, permission, export and destructive actions.
- Deactivate accounts immediately when staff leave.

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Lead and CRM workflow

The CRM can remain lightweight at launch while still providing ownership, history and measurable conversion.

```
NEW -> CONTACTED -> QUALIFIED -> QUOTED -> WON
      -> LOST
      -> ON HOLD
```

Capability	Launch requirement
Assignment	Owner or sales user responsible for next action
Activity	Status changes, notes, calls, emails and timestamps
Source	Project, service, package, campaign and landing page
Follow-up	Next-follow-up date and overdue dashboard indicator
Quotation	Reference/link and quotation value; full quotation engine can remain later
Conversion	Create Client and operational Project without retyping data
Export	Permission-controlled CSV with audit event

SCOPE CONTROL

Do not build a large enterprise CRM in version 1. Start with reliable lead capture, status, ownership, notes, attribution and follow-up.

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SEO implementation

SEO must be built into the content model and rendering strategy rather than added as a plugin after launch.

Area	Implementation
Rendering	SSR/SSG/ISR for public content; meaningful HTML without JavaScript
Metadata	Unique title, description, canonical, social image and index control
URLs	Stable human-readable slugs and managed 301 redirects
Sitemap	Automatic published routes; exclude drafts, filters and admin
Structured data	Organization/LocalBusiness, Service, VideoObject, Breadcrumb and valid Review
Video	Poster, title, description, duration, upload date, captions and transcript where useful
Local	Service-area pages using genuine content; consistent business contact details
Internal links	Projects ↔ services ↔ related work ↔ enquiry

CMS validation

- Warn when title or description is missing/duplicated.
- Generate a safe slug but allow controlled editing before first publish.
- Require alt text for informative images.
- Preview search and social snippets.
- Never generate fake location pages or keyword-stuffed paragraphs.

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Accessibility specification

Target WCAG 2.2 AA and test real interactions—not only automated color checks.

Concern	Requirement
Structure	One H1, logical heading order, landmarks, skip link and meaningful page titles
Keyboard	Menus, filters, modal, players, slider, carousel and forms fully operable
Focus	Visible focus; trapped/restored focus in modal; no focus hidden behind sticky UI
Media	Captions for meaningful speech, transcript when valuable, no forced audio
Images	Useful alt text; decorative imagery ignored; logo names available
Motion	prefers-reduced-motion; no flashing; pause controls where needed
Forms	Persistent labels, instructions, field errors and error summary
Touch	Minimum 44×44px target and adequate spacing
Contrast	4.5:1 normal text; 3:1 large text and UI components

MANUAL TESTS

Complete the entire enquiry with keyboard only, screen-reader-test the page structure, zoom to 200%, test reduced motion and verify captions with sound disabled.

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Performance budgets

A cinematic site must feel immediate before it starts playing video. Poster-first delivery is mandatory.

Metric / asset	Target
LCP	$\leq 2.5s$ at the 75th percentile
INP	$\leq 200ms$ at the 75th percentile
CLS	≤ 0.1
Initial JavaScript	Preferably below 200KB compressed; justify every client library
Hero image	Responsive AVIF/WebP; do not load full video before consent/intent
Fonts	Subset/self-host where licensed; preload only critical files
Gallery	Responsive srcset, lazy loading, explicit dimensions
Video	Stream adaptive formats; lightweight poster; lazy-load player SDK

Implementation tactics

- Use server components for content and small client islands for interaction.
- Reserve media aspect ratios to prevent layout shift.
- Cache public APIs and invalidate by content tag.
- Do not autoplay multiple reels in a scrolling grid.
- Monitor real-user performance after launch, not only local Lighthouse scores.

23 / IMPLEMENTATION BLUEPRINT

Security, privacy and rights

The public form, admin access and media library introduce real business and privacy risk.

Risk	Control
Admin compromise	MFA, strong session settings, rate limits, audit log and least privilege
Upload abuse	Signed short-lived uploads, signature validation, size limits and processing isolation
Spam	Server validation, honeypot/bot protection, rate limiting and alerting
Injection/XSS	Parameterized EF queries, output encoding, sanitized rich text and CSP
Secret exposure	Managed secrets; never ship storage/API credentials to client JavaScript
Personal data	Collect only needed lead data; retention and access/delete process
Media rights	Rights entity, consent/license evidence, usage scope and expiry reminders
Webhook forgery	Verify signature, timestamp and idempotency
Data loss	Automated database backups and tested restoration

LEGAL REVIEW

Privacy, cookie, event-filming consent, music licensing, drone permission and testimonial/logo usage require business/legal confirmation before launch.

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Analytics and business measurement

Track only events that answer a product or sales question. Avoid collecting unnecessary personal data.

Question	Events / dimensions
Does the hero work?	hero_cta_clicked by action and device
Is the work relevant?	project_opened by event type and source section
Do visitors watch?	showreel_started, 25/50/75%, completed
Which scope converts?	package_selected and enquiry_started with packageld
Where do leads come from?	enquiry_submitted with sourceProjectId/serviceld/campaign
Where does the form fail?	validation summary without recording sensitive values
Is content current?	publish frequency, media failures and stale featured content

Dashboard metrics

- Qualified enquiries per month and by source.
- Enquiry-to-contact and contact-to-win conversion.
- Most-viewed project types and services.
- Showreel completion rate.
- Median response time.
- Do not treat total page views as the primary business outcome.

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Testing strategy

Test the business flow and content growth—not only individual functions.

Layer	Coverage
Unit	Validation, slug rules, pricing labels, publish eligibility, permissions and status transitions
API integration	Database writes, idempotency, concurrency, public filtering and webhook handling
Frontend component	States, accessibility, long content, keyboard and responsive behavior
End-to-end	Browse → select package/project → submit enquiry → admin sees lead
Publishing	Draft → preview → publish → revalidate → unpublish
Media	Upload, failure, retry, poster, caption and archive/purge
Visual regression	Core pages at mobile/tablet/desktop breakpoints
Performance	Lighthouse CI plus production real-user metrics
Security	Auth/session, permission bypass, upload validation and rate-limit tests

Critical end-to-end scenarios

- Add the 11th client and confirm layouts remain stable.
- Add a 12-person team and verify homepage/full-page limits.
- Publish ten reels and verify pagination, playback and performance.
- Submit the same enquiry twice with one idempotency key and store one record.
- Expire a media right and prevent publication or surface a warning.

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Deployment and operations

Keep public delivery, API, database and media independent so each layer can scale or recover without taking everything offline.

Component	Deployment responsibility
Next.js	Managed frontend platform or container; CDN; preview deployments
ASP.NET API	Containerized service with health checks and autoscaling policy
PostgreSQL	Managed database, private networking, PITR/backups
R2 / Stream	Buckets, lifecycle, CORS, signed access and webhook configuration
Background jobs	Outbox/queue worker for media, email and cleanup
Observability	Structured logs, traces, error alerts and uptime monitoring

CI/CD gates

```
Pull request -> lint/typecheck -> unit/integration tests
-> accessibility/visual checks -> preview environment
-> approval -> database migration guard -> production release
-> smoke tests -> monitor -> rollback if required
```

- Use development, staging and production environments.
- Migrations are forward-reviewed and backed up before risky changes.
- Content publication remains independent from code deployment.
- Document recovery for database, API, frontend and media separately.

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Suggested repository structure

A monorepo keeps contracts and workflows coordinated while preserving clear boundaries.

```
fix-frame/  
  apps/  
    web/           # Next.js public website  
    admin/         # protected React/Next.js admin  
    api/           # ASP.NET Core API  
    worker/        # media/email/outbox jobs  
  packages/  
    ui/            # design-system components  
    contracts/     # generated TS API types  
    config/        # lint, TypeScript, tokens  
  infrastructure/  
    containers/  
    migrations/  
    deployment/  
  docs/  
    architecture/  
    content-model/  
    runbooks/  
  tests/  
    e2e/  
    visual/
```

ASP.NET boundaries

- Domain: entities, value objects and business rules.
- Application: commands/queries, validators, permissions and interfaces.
- Infrastructure: EF Core, R2/Stream, email, jobs and persistence.
- API: controllers/endpoints, authentication, middleware and contracts.
- Prefer feature-based folders inside each layer once the project grows.

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Step-by-step implementation plan

Build vertical slices that produce a usable product early. Do not finish every admin screen before validating the public experience.

Phase	Deliverable	Exit condition
0. Discovery	Real content inventory, rights, copy, domains and success metrics	No critical placeholder dependency
1. Foundations	Repos, environments, auth, DB, tokens and component base	CI green; preview available
2. Public shell	Header/footer, hero, showreel, responsive layout	Core accessibility and performance pass
3. Content vertical	Projects + media + admin + publish/revalidate	Owner publishes first case study
4. Collections	Services, packages, reels, team, clients and testimonials	All growth rules tested
5. Enquiries	Form, storage, notifications, CRM status and attribution	End-to-end lead test passes
6. SEO/all	Metadata, sitemap, schema, captions and manual testing	Audit issues resolved
7. Operations	Backups, monitoring, runbooks, analytics and security review	Launch checklist approved
8. Launch	Production migration, smoke tests and observation	Stable production release

DEVELOPMENT ORDER

The first complete vertical slice should be Project + Media + Admin + Public case study + Publish. It proves the hardest dynamic-content workflow before the remaining collections are added.

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Definition of done

The project is complete when the owner can operate it safely and the public site performs as a credible business tool.

- ☐ Approved public sections are implemented with real Fix Frame content.
- ☐ Visual system is consistent and does not resemble a generic AI template.
- ☐ Owner can create, preview, publish, reorder, unpublish and archive content.
- ☐ Adding 10+ reels, clients and team members requires no code or deployment.
- ☐ Homepage feature limits and full-page pagination work correctly.
- ☐ Original media is retained; derivatives and Stream playback are reliable.
- ☐ Structured enquiry is validated, stored, notified and visible in CRM.
- ☐ packageId, serviceId and sourceProjectId attribution is preserved.
- ☐ SEO metadata, sitemap, schema, redirects and social previews are functional.
- ☐ WCAG 2.2 AA critical flows pass automated and manual tests.
- ☐ Core Web Vitals and media performance meet agreed budgets.
- ☐ Permissions, MFA, audit, rate limits, backups and rights controls are verified.
- ☐ Staging, production, monitoring, rollback and recovery procedures are documented.
- ☐ The client receives admin training and a short operating guide.

FINAL STANDARD

The website should remain professional when the content changes, the team grows and the portfolio expands. That maintainability is part of the design—not a separate backend concern.

FIX FRAME / BUILD PRINCIPLE

**PROTECT THE DESIGN.
STRUCTURE THE CONTENT.
AUTOMATE THE GROWTH.**

PUBLIC EXPERIENCE	OWNER CONTROL	BUSINESS OUTCOME
Fast, cinematic, readable and accessible	Dynamic content without normal redeployment	Qualified enquiries and credible proof

Use this blueprint as the shared implementation contract for design, frontend, backend, content preparation and launch.